TRUSTMYBIKE

Smart Bike for Road Quality Mapping

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Start

PROJECT OVERVIEW

Our goal is to produce a smart bike that evaluates road quality and creates shareable maps for the cycling community. TrustMyBike helps bikers choose safer, smoother routes by understanding road conditions in advance. The key concept combines real-time road quality evaluation with collaborative map sharing across devices, enabling cyclists everywhere to benefit from crowdsourced road condition data and make informed decisions about their routes.

SYSTEM ARCHITECTURE

Sensors

MPU6050 accelerometer detects holes and measures speed. Gyroscope tracks street slopeness. Phone GPS provides location data.

Power Source

Bike wheel dynamo generates 6V, 3W max output, producing up to 500 mA of current.

Power Consumption

Accelerometer: 500 µA, Gyroscope: 3.6 mA, Bluetooth: 130 mA, ESP: 50 mA. Total: 187.1 mA (within capacity).

SAMPLING & LEARNING

1 Sampling Frequency

Formula: F = 2 × speed / bump length (based on Sampling theorem). Frequency depends on bike speed and bump size. With medium wheel circumference of 70 cm and bump defined as 10 cm diameter.

2 Machine Learning

Classify street quality from sensor data collected by accelerometer and gyroscope. Detect hazards including holes and uneven slopes. Improves route safety and overall user experience.

TrustMyBike

COMMUNICATION FLOW & SHARED MAP

Data Path:
MPU6050 sensors → ESP microcontroller (classifies street quality) ↔ Phone (enriches data with GPS coordinates) → Cloud (uploads and builds shared map)

Outcome:

- Collaborative map accessible to all bikers worldwide
- Real-time updates on road conditions and hazards
- Community-driven route optimization
- Safer cycling through shared knowledge

THANK YOU!